

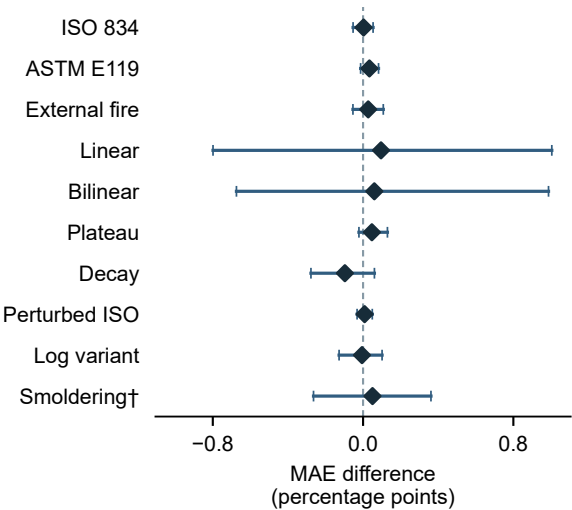
a Generalization to each completely held-out fire family

All ten families · seven models · five seeds · 350 complete audited runs

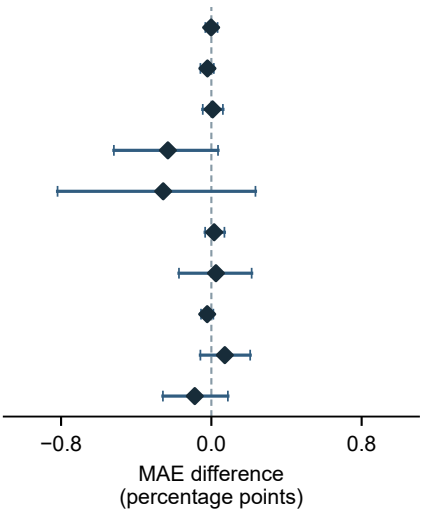
	Fire only	Global stats	DeepSets	Matched DeepSets	GNN sum	GNN zero-edge	GNN mean	MAE (pp)
ISO 834 (158)	1.181	0.894	0.838	0.840	0.837	0.846	0.838	4
ASTM E119 (95)	1.198	0.979	0.813	0.818	0.784	0.819	0.805	
External fire (49)	1.599	1.238	1.090	1.053	1.025	1.038	1.018	
Linear (125)	4.043	3.257	3.782	2.629	2.534	2.602	2.765	
Bilinear (77)	3.642	2.748	3.036	3.160	3.100	3.590	3.357	3
Plateau (103)	1.359	1.139	0.993	0.980	0.933	0.968	0.918	
Decay (113)	1.537	1.252	1.366	1.386	1.483	1.338	1.458	
Perturbed ISO (111)	1.216	0.957	0.880	0.881	0.873	0.893	0.895	
Log variant (69)	1.661	1.434	1.341	1.377	1.383	1.362	1.311	2
Smoldering† (97)	0.726	1.073	1.028	1.211	1.160	1.235	1.249	
Case-weighted (997)	1.804	1.486	1.521	1.406	1.385	1.432	1.437	
Equal-family (10)	1.816	1.497	1.517	1.433	1.411	1.469	1.461	

b Matched DeepSets – sum GNN

Prespecified paired differences: 95% geometry-and-seed intervals; positive favors the second model



c Sum GNN – mean GNN



† Smoldering: 86/97 cases are restored flat responses; 11 retained cases are nonflat.

Intervals are unadjusted; the two cross-family averages are descriptive, with no pooled interval.